Sai Kiran Goud Lingala

CFD and Thermal Engineer | Simulation, Heat Transfer and Model Validation
Indianapolis, IN | (937) 516-4650 | saikiran27.gl@gmail.com
saikiran4099.github.io | linkedin.com/in/saikiranlingala27 | github.com/SaiKiran4099

SUMMARY

M.S. Mechanical Engineer specializing in computational fluid dynamics and thermal analysis. Build and validate simulation models against published experimental and benchmark data, diagnose discrepancies to root cause rather than tuning until the number looks right, and automate analysis workflows in Python. Experience delivering analysis as client-facing engineering work and teaching heat transfer at university level. Publish validation studies openly, including the cases that failed first.

SKILLS

CFD and Thermal Simulation: Simcenter STAR-CCM+, ANSYS Fluent, OpenFOAM, ANSYS Meshing, SpaceClaim, ParaView; steady and transient analysis, conjugate heat transfer with coupled solid and fluid domains, RANS turbulence modeling (k-omega SST, Realizable k-epsilon), moving reference frame and sliding-mesh rotating domains, multi-region interfaces, compressible and incompressible flow

Heat Transfer and Fluid Mechanics: conduction, convection and radiation, thermal resistance and heat flux path analysis, wall heat flux and surface temperature prediction, cooling passage and coolant flow path analysis, airflow distribution and pressure drop, boundary layers, separation and reattachment, thermodynamics, rotating machinery, aerodynamics; turbulence closure benchmarking against experimental and DNS data

Verification and Validation: correlation against published experimental and DNS reference data, GCI and Richardson extrapolation, mesh and solution independence, uncertainty quantification, mesh quality verification, signed-deviation reporting against acceptance criteria, root cause diagnosis of model discrepancies

Meshing and CAD: trimmed, structured and hybrid meshing, prism and boundary-layer control to y+ of 1 or below, geometry cleanup, defeaturing and watertight body preparation for simulation; SolidWorks, CATIA V5/V6, ANSYS SpaceClaim; ANSYS Workbench static structural

Programming and Tools: Python (NumPy, Pandas, Matplotlib, PyTorch), MATLAB, solver journal scripting and batch workflow automation, Git version control, Linux and Windows, Microsoft Office

EXPERIENCE

CFD and Thermal Analysis Engineer Jan 2022 – Jul 2023
Cyient, Hyderabad, India

- Owned thermal-fluid analysis end to end on a client hardware programme, from CAD geometry preparation and meshing through solution, post-processing, interpretation and written design recommendation.
- Ran a 14-case parametric campaign, diagnosed a flow separation and performance loss to root cause, proposed a geometry design change, and verified 7 percent CL and 11 percent L/D improvement against the pre-change baseline.
- Established acceptance criteria and held validated results within 0.8 percent of reference data using GCI and Richardson extrapolation across three mesh densities.
- Automated boundary-condition setup, multi-case data reduction and report generation in Python and solver journal files, cutting turnaround roughly 35 percent and removing manual transcription error.
- Reported status, open issues and recommendations to cross-functional teams across propulsion, controls and systems integration, and to client stakeholders without a simulation background.

Graduate Teaching Assistant, Heat Transfer and Controls Aug 2023 – Aug 2025
University of Dayton, Dayton, OH – taught conduction, convection, radiation, thermal boundary layers and internal flow at undergraduate and graduate level; mentored 20+ students and gave written technical feedback on analysis work.

SELECTED WORK – published at saikiran4099.github.io

Root Cause Diagnosis of a Model Failing Its Benchmark Simcenter STAR-CCM+

- Built a rotating machine two independent ways, steady moving reference frame and transient sliding mesh, on a 774,069-cell two-region trimmed mesh; both over-predicted the published benchmark by a factor of 2.1.
- Tested and refuted rotor-stator unsteadiness and turbulence-model bias before a dimensional audit isolated a 25 percent oversized swept volume; predicted the corrected result analytically before rebuilding, then confirmed within 3 percent of benchmark and 0.74 percent between formulations.

Conjugate Heat Transfer and Cooling Configuration Trade-Off ANSYS Fluent

- Coupled solid and fluid domains on a cooled high-temperature component so conduction and convection resolved together rather than assuming a heat transfer coefficient at the interface, extracting wall heat flux and surface temperature per configuration to support a design recommendation.

EDUCATION

University of Dayton, Dayton, OH M.S. Mechanical Engineering, Dec 2025
Concentration: heat transfer, computational fluid dynamics, compressible flows, propulsion

MLR Institute of Technology, Hyderabad, India B.Tech. Aeronautical Engineering, Jul 2022
FE Mechanical exam scheduled September 2026. Co-inventor, filed Indian patent application – morphing wing mechanism varying thickness-to-chord ratio in flight.

Work Authorization: F-1 OPT through Feb 2027, eligible for the 24-month STEM OPT extension: authorized full-time work with no petition and no employer filing beyond E-Verify enrollment and a signed I-983. Open to relocation.